

Wenyin (Knut) Gu

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EDUCATION

- **University of California, Irvine** | Irvine, CA | 2025 - Present
B.S. in Electrical Engineering (Specialization: Digital Signal Processing)
GPA: 3.69/4.0
- **Cañada College** | Redwood City, CA | 2022 - 2025
A.S. in Engineering
GPA: 4.0/4.0

EXPERIENCE

- **Fundamentals of Data Science Workshop**, *Stanford University* | Summer 2023
Completed a 4-week intensive program encompassing Applications of Data Science, Python, Statistics, Mathematical Optimization, Linear Algebra, and Responsible AI.
- **Tutor of Mathematics**, *Cañada College* | 2022 - 2024
Provided tutoring at the Learning Centre, guiding students through mathematical problems and supporting their development of analytical skills.
- **Outstanding Service Contributor (Water Supply Team Lead)**, *Shanghai World Expo* | 2010
Conducted comprehensive inspections of water supply risks surrounding the venue, evaluated safety measures, and enforced administrative penalties.
- **Volunteer**, *Special Olympics World Summer Games* | 2007
Provided dedicated care and round-the-clock support for the football team's daily routines, ensuring their safety and well-being during training.

PROJECTS

- **Music Reactive LED System** | *Electrical Engineering Project*
Designed a custom microphone preamplifier and a 3-band active analog crossover network using LM358 Op-Amps to separate audio into Bass, Midrange, and Treble channels prior to digitization.
Programmed a non-blocking state machine on an Arduino Mega to process ADC sampling into three distinct LED visualization modes: an analog-driven Spectrum Analyzer, a real-time OLED Waveform Monitor, and a dynamic Reactive Strobe.
Integrated a 0.96-inch I2C OLED display and a rotary encoder to build an interactive GUI, enabling seamless mode switching and real-time gain control without modifying code.

SKILLS

- **Software & Programming:** Python, MATLAB, SolidWorks, Onshape, C/C++ (Arduino).
- **Hardware & Electronics:** Analog Circuit Design (Op-Amps, Active Filters), Microcontroller Integration (ADC, I2C), UI System Implementation, Soldering/Perfboard Fabrication.

AWARDS and HONOURS

- IEEE-Eta Kappa Nu (HKN) | 2026
- Phi Theta Kappa | 2023
- Valedictorian & *Summa Cum Laude*, Cañada College
- Dean's List, UC Irvine & Cañada College